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## Digital divides: territory, gender, age, class, ethnicity, cultures

The digitalization of society has spread globally at a fast pace, and it is projected to expand further in the short term (2030), as I documented in the Introduction to this volume. However, the diffusion of technological affordances and users' access and capabilities vary in step with the sources of inequality around the world. Moreover, the United Nations Agenda 2030 redefined its goals for digitalization from aiming for "universal access" to "universal and *meaningful* access," a distinction that we aim to make more precise when examining the empirical record in this chapter.<sup>1</sup>

### Territorial divides

In the age of globalization and Internet networking, digital inequality between countries remains, largely associated with the wealth of the country. The percentage of Internet users over the global population in 2022 was 66%, but it reached 92% in the high-income countries, 79% in the upper-middle-income countries, falling to 56% for the lower-middle-income group, and to 26% for the low-income countries. There is also a difference of penetration for smartphones, although there is a high proportion of users of smartphones everywhere, the highest percentage being in Europe, with 92%, the same percentage as in the US. Mobile cellular subscriptions reach 86% in Africa and over 100% in most regions of the world, with 121% the figure in Europe, illustrating the unequal distribution, whereby some people have several subscriptions while others, even if they are a small minority, have none. Territorial

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<sup>1</sup> For simplicity, references for statistics on ICT use will be omitted in the text. They are included in the references section, and are typically from ITU when it comes to aggregates. Exceptionally, data is sourced from Statista or the Pew Research Center.

inequality extends, in every country, to the sharp disparity between urban and rural areas, while the need for broadband connectivity is more acute for rural localities. Indeed, a study by Galperin et al. (2022) on rural Ecuador in 2011–19 illustrates this. The researchers found substantial gains in income, and a modest improvement in employment, associated with the introduction of high-speed broadband that facilitated access to critical market information, and helped to diversify employment. In a larger context, a statistical analysis of 24 Latin American countries in 2020 calculated that an increase of 1% in fixed broadband penetration yields an increase of 0.08% of GDP, and a 1% increase in mobile connectivity increases GDP by 0.15% (Ziegler et al., 2020). And yet, in Latin America, where rural poverty is double that of urban poverty, Internet penetration in rural areas stood at 37% in 2020 in contrast to 71% in urban areas. However, this is somewhat compensated by the penetration of smartphones in rural areas, which reached 71% in 2020 (Ziegler et al., 2020). Worldwide, while urban areas displayed an 82% Internet penetration in 2022, rural areas lagged behind at 46%. The rural–urban cleavage persists everywhere, but is less pronounced in highly developed countries: in the US in 2021, Internet use in urban areas was 95%, and in rural areas 90% (Statista, 2021). In the UK in 2020, Internet penetration in urban areas was 98% and in rural areas 94% (Statista, 2020a; 2020b), although here again mobile phone penetration at 130% ensures communication in most of the country. China presents an intermediate situation, with 77% urban penetration and 46% rural access to the Internet in 2020 (ESGN, 2021). In all cases the gap in quality of the connection is substantial, as the rural infrastructure lags behind the networks deployed in the main urban centers.

There is a second territorial divide, namely the location of the infrastructure for digital communication, tantamount to a differential access to the “cloud,” because the cloud materializes in servers that store and direct data traffic originated anywhere. As of 2018, the large majority of international traffic from Latin American countries went to servers in North America (73%) and Europe (10%). In the case of African countries, 56% of the international traffic went to North American servers and 32% to Europe. Even Europe is not self-sufficient in its servers, as 43% of traffic used North American servers, while 56% of international traffic went to another European country. China relies on its own servers for domestic traffic, but its global access may also be dependent on North American servers that manage 49% of international traffic from Asia–Pacific, which

includes Chinese data. The hubs of international traffic are located in a few areas to serve entire regions of the world: four of the five highest capacity international routes in Latin America are connected in Miami (Telegeography, 2018). The disparities in infrastructure location and control are accentuated by the ownership of exclusive undersea cables by major digital companies such as Google, Facebook, Amazon, and Microsoft that account for the largest share of the increase in global bandwidth in recent years (Satariano, 2019). Preferential location to the hubs of the global infrastructure has substantial consequences for the design of national technology policies, and obvious implications for national security, thus decreasing sovereignty in the majority of countries as we transition to a more digital environment (Ortiz Freuler, 2020).

The deployment of advanced telecommunications infrastructure is also differential between countries, as illustrated by the global distribution of 5G networks (Statista, 2023). With the significant exception of sub-Saharan Africa, by 2020 most countries had launched, deployed, or invested in 5G. For instance, most of South America had already launched the networks and the remaining countries had deployed the technology. China is among the most developed users of 5G because Huawei was critical in the design of the technology, and set up 70% of the base stations in the world as of 2023 (Parzyan, 2023). But Indonesia also is in the leading group while Russia is in an earlier phase of development. The geography of telecom networks is a major indicator of the extent of inequality in the process of digitalization worldwide.

## The gender gap

The gender gap in the digital world goes beyond the difference between men and women in Internet access. Indeed, in 2022, the gap was limited in terms of global statistics: 70% of men were daily users, a somewhat higher proportion than 63% of women. But this difference masks substantial variation among countries, depending on the geography of patriarchy. In the UK in 2020, men were more likely to use the Internet daily (90% versus 89%; ONS, 2021), while the figures reported for the US in 2023 were similar (94% for men and 93% for women). In Spain in 2021, the percentages of users were close (77% for men, 75% for women) (Garín-Muñoz et al., 2022). But the differences are greater in certain

regions of the world. The proportions of Internet use for men and for women were respectively 83% and 83% for the Americas, 89% and 90% in Europe. However, the gap was larger in other regions, with 67% and 61% in Asia–Pacific, 75% and 65% in the Arab States, and 46% and 34% in Africa, respectively.

Furthermore, the world's Internet use among women in high-income countries was 92%, in sharp contrast with the low-income countries where only 21% of women use the Internet. Even in China, now a rather developed country widely using mobile phones, in 2020 74% of men owned smartphones but only 54% of women did. In sum, the level of development of countries specifies the digital divide: the more developed the country is, the greater the trend toward shrinking the gender gap. And vice versa. This is clearly related to the condition of women in most of the Global South, widely submitted to patriarchal domination.

However, access is not the only dimension of the gender gap in the digital age. Pay in the digital industries is substantially lower for women, controlling by the type of occupation: it is 20% lower than men in the US, and 33% lower in less developed countries (ILO, 2019). The leading companies in the digital industry employ a minority of women: only 17% of Facebook's employees are women, 19% for Google, 23% for Apple (Chakravorti, 2017). This is largely the result of the low proportion of women graduating in science, math, engineering, and technology, about 25% of all graduates in the US. One of the leading researchers in this field, Cecilia Castano has thoroughly examined in a series of studies the roots of this low presence of women in technology occupations, and found it in the triage operated in high schools toward learning for gendered labor markets, and in the discriminatory culture prevalent in a number of engineering schools (Castano, 2005a, 2005b, 2015, 2021; Castano et al., 2010; Castano & Webster, 2011, 2014).

Furthermore, women attempting entrepreneurship in the digital industry face considerable difficulties. According to OECD (2018) studies, women receive 23% less funding than their male counterparts to start a company. There are more subtle, and more damaging, differences in gender access to social media. In general, around the world, more women than men use social media. An internal study commissioned by Instagram found that social media made teens feel worse about their bodies and that they blame the platform for anxiety, depression, and suicidal thoughts (Wetsman,



2021). Some studies in the UK have shown that young girls (11–13) are facing greater emotional challenges than their male classmates (Fink et al., 2015). Large-scale surveys in the US report teen girls being subjected to online bullying and aggression more often than their male counterparts (Pew Research, 2022). Meanwhile, a growing number of studies support the association between loneliness, depression, and suicidal thoughts with the use of smartphones and social media (Shensa et al., 2018; Allcott et al., 2020; Twenge, 2023). Furthermore, the deficit in the presence of women in digital design (e.g., video games) tilts the content toward a predominantly male, and often violent, culture that appeals to young high-testosterone males and excludes other habits and visions of the world produced by women's specific experience. In my conversation in 2019 in Shenzhen with Mr. Ma, founder and CEO of Tencent, the Chinese company with the largest market share of video games in the world, he expressed openly his concern for this gender bias, and his desire to gradually change the content of the games.

In sum, the digital gender gap and its unwanted consequences result from the differential levels of development of societies, the systemic discrimination against females, their cultural apartheid in certain roles and occupations, and the still sharply contrasted geography of patriarchy, with its multiple sexist corollaries.

## **The grey divide**

Differential use, access, and quality of Internet connection depending on age is the most significant divide in this early digital society, because it is likely that as the people in the age cohorts over 65 and 75 years, who grew up before the advent of the Internet, pass away, the overall digital divide in terms of access will dwindle. Indeed, the proportion of the world population over 60 years old in 2021 was 12%, while the proportion of persons older than 65 in the population of Internet users was 6% (Statista, 2021). On the other hand, the proportion of the population under 20 years is 33%, while the percentage of Internet users in the group 25–34 is about one-third of all Internet users, and in the group 18–24 was 18%. However, this demographic trend does not imply that the digital divide will disappear by biological law, because in 2022, while the percent of Internet users in the age group 15–24 reached almost 100% in the high-income

countries, in the low-income countries it was still under 40%. The wealth of countries and the uneven distribution of technological capacity will still translate into a digital divide for the years to come. The situation is increasingly different in developed countries, yet a grey divide still exists. Thus, in the US, in 2021 in the group 18–29 years old, 99% are Internet users, while among 65 years and older, the proportion is 75%.

Indeed, a significant number of seniors have not joined the digital age: in the UK, 26% of people 65–74 and 61% of those older than 75, do not regularly use the Internet. In the US, 25% of people older than 65 are not Internet users. In the European Union, the use of the Internet, in spite of its availability, varies considerably for the age group 65–74, from 90% in Denmark and 70% in Spain, to 40% in Portugal and 25% in Croatia. Furthermore, the stream of studies conducted over the years by Mireia Fernández-Ardèvol and her international network of researchers have questioned the linearity of the relationship between age, use, and meaning of mobile Internet for the older groups of the population (Fernández-Ardèvol et al., 2017; 2023; Fernández-Ardèvol, 2020; Beneito-Montagut et al., 2022). They have shown the key role of smartphones in the communicative practices of the elderly, mainly because of the multimodality of the devices. On the basis of qualitative studies, they emphasized what they call techno-diversities, in terms of the use of communication devices over the life cycle, particularly the significant change of use and meaning after 75 years, what they call the “fourth age” group. For many elderly, the nonuse or the selective use of Internet and smartphones is not necessarily due to lack of digital capabilities, but to a mechanism of empowerment, by deciding what, with whom, and how to communicate. In other words, the inclusion/exclusion in the networks of digital communication is not the result of unilateral policies of companies (such as the pricing out of low-income elderlies) but of an interaction between life histories, and adaptation to a disadvantaged position, worsened by hyper digitization, ultimately resulting in the affirmation of the right to decide their own life, in spite of physical and cultural disabilities. The right to be left alone, although not excluded from society, is becoming the new frontier of digital rights, beyond the strategies and interests to expand markets for telecommunication business.

## Digital divide by class

Class, as usually defined in stratification analysis, refers to the differentiation of people in terms of their wealth (income, assets), level of education, and occupational status. In fact, these three categories correlate. In all countries, there is a direct relationship between the high ranking in each one of these categories, and access to and use of the Internet (Mubarak et al., 2020). Moreover, even in contexts where social hierarchies are defined in cultural and institutional terms, as in the case of caste in Indian society, income and education remain the key factors in explaining the caste stratification in Internet use (Rajam et al., 2021).

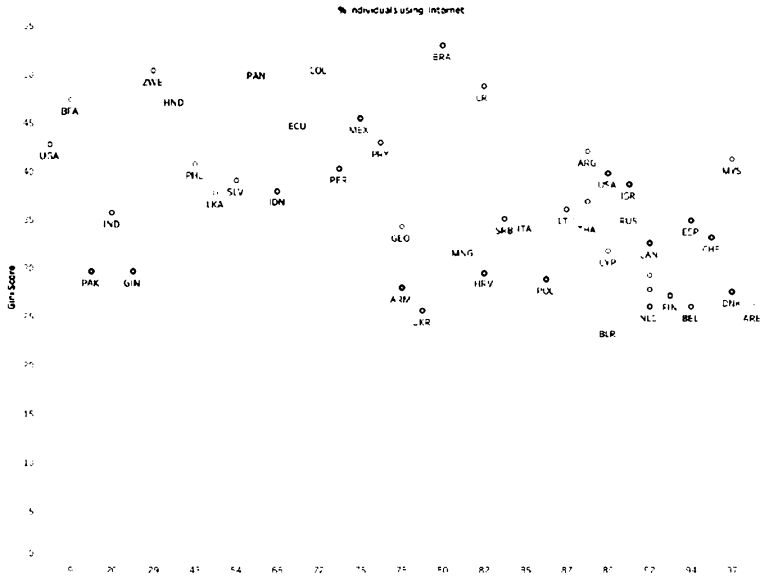
The educational background of the family reproduces the social differentiation of their children in their Internet access, both at home and at school. In the US, for instance, in 2021, 98% of children in the group 13–18 years old with parents with college education or higher have home Internet access, while among those children with parents with an educational level of high school, 79% have home Internet access (NCES, 2021). In low-income countries, the difference in access according to family background is much larger.

However, as smartphone ownership spreads around the world, it is necessary to update the understanding of digital inequality. Indeed, in 2023, 86% of people in the world had a smartphone, and 91% had a mobile phone. Considering that 94% of global Internet users accessed the Internet from their smartphones (it should be noted that the region where the proportion of access to the Internet by mobile/smartphone is highest is Africa), statistics based on home or school access to the Internet have to be complemented by including the use of smartphones. Granted, however, that learning by smartphones offers, in theory, fewer possibilities than using a dedicated school computer. Yet, in the US in 2022, 55% of high school students used their smartphones for schoolwork. Since the technological affordances of smartphones grow exponentially, the real divide may be less in access to the Internet from home or school than access to broadband and quality (and price) of smartphone technology, therefore increasing the relevance of income, education, and infrastructure in the construction of digital inequality.

In sum, since inequality in the distribution of resources (be it economic or cultural) is prevalent around the world, and increasingly so (Piketty

& Rendall, 2022), it is not surprising that inequality is also a defining feature of the distribution of digital communication use and capacity at the global level. However, the question remains of the extent of digital inequality compared to other sources of inequality. There is scant quantitative research on the matter. Nonetheless, one indication in this regard comes from the study I directed on the diffusion, uses, and impact of mobile communication in Latin America (Fernández-Ardèvol et al., 2011). Comparing inequality between countries according to income and inequality according to the penetration of mobile communication, we observed that income inequality was more pronounced than mobile communication inequality. We ventured some hypothetical explanation for the reasons for the spread of mobile telephony throughout all of the social groups using ethnographic analysis. While higher-income groups obviously benefitted from better equipment and infrastructure, most low-income people (including the urban poor and farmers) felt the importance of using mobile communication, both for their working needs and their personal lives, including to keep emotional and financial ties with their family migrants.

Furthermore, a well-established literature has provided evidence of the correlation between the deployment of digital infrastructure and skills level, and the economic growth of countries (Vu et al., 2020). While it may appear that the most pressing needs are to remedy the lack of electricity, water, amenities, and public services, the capacity to finance those social investments depends on economic growth that, in the current global economy, is partly a function of the technological modernization of economies and societies. Ortiz Freuler has analyzed the relationship between inequality in countries and their level of Internet access, which shows a strong relationship between inequality levels as measured by the Gini index and Internet access between countries, where the lower the inequality, the higher the Internet access (see Figure 7.1). In-depth research analyzing 86 countries has found that results are varied across countries, but within low-income countries an increase in Internet access is associated with a reduction in poverty, and access to broadband with a reduction in income inequality (Afzal et al., 2022).



Source: By Ortiz Freuler using ITU data.

**Figure 7.1** Scatterplot comparing Internet use and Gini score

## Ethnicity divide

It is generally assumed that ethnic minorities are disadvantaged in terms of Internet access. We do not really know because many countries (France and Germany, for instance) do not include race or ethnicity in their Census categories, a dubious symbol of calling for citizens' equality beyond ethnicity. The US does have ethnically differentiated statistics, both in the Census and in a number of surveys (US Census Bureau, 2021). The story the data tell is nuanced. Counting the proportion of Internet users within each ethnic group, in 2019, the highest group, at 99%, were the Chinese-Americans. In 2021, a survey of adults of a representative sample of recent users of the Internet published by Pew showed some unexpected findings: while 93% of whites were users of the Internet, Hispanics had a higher proportion (95%) with African Americans not too far behind at 91% (Pew Research, 2021a). On the other hand, in 2021, there were significant differences in terms of home broadband: 80% of white homes have access, in contrast to 71% of African Americans, and

67% for Hispanics. One possible explanation for this discrepancy could be that the proportion of ownership of smartphones (with Internet access potential) was very similar: 85% for whites, 85% for Hispanics, and 83% for African Americans (Pew Research, 2021b). Much of the perceived gap between whites and other ethnic minorities results from differences in income, education, territory, and lack of infrastructure that disproportionately hit ethnic minorities. Rural African Americans, particularly in the Southern states, were nearly twice as often excluded than white Americans in the same areas (Asher-Schapiro & Sherfinski, 2021).

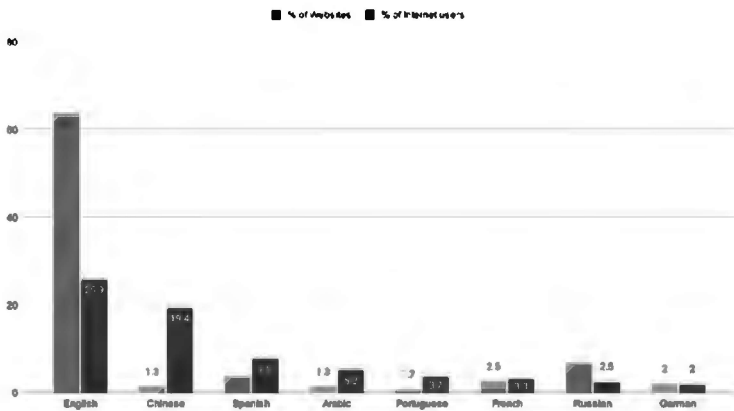
A new form of ethnic discrimination has appeared in the digital environment: some social media platforms allow their advertisers to discriminate against groups protected by civil rights laws. After the intervention of the Department of Justice, in 2019 Facebook settled a lawsuit by civil rights groups by creating a special portal for ads related to credit, housing, and employment, to ensure targeting techniques would not be used to discriminate against disadvantaged groups (Tobin & Kofman, 2022). This is probably only the tip of the iceberg of a new form of digital discrimination.

In sum, historically disadvantaged ethnic minorities suffer from a multiplicity of sources of inequality (income, assets, education, infrastructure) whose combined effect challenges the optimistic notion of equalization of opportunities in the digital age.

## Cultural divides

We live in a multicultural world. And yet, while cultures are all born equal, history has not treated them equally, both within each country and in the global realm. Cultures are codified in language. Although language is not the only component of a given culture, it is the decisive one because it is what allows communication among those identifying with the culture. And yet, when we express ourselves and communicate in the global agora, which is what the Internet is supposed to be, most people have to use a language, thus a cultural construct, that is not theirs. Indeed, English is the Internet's universal language (Statista, 2022). According to UN statistics, about 1.5 billion people speak English (this of course includes part of the population of India), of which 1.2 billion are Internet

users. This represents approximately 30% of the world's Internet users. A critical observation is that 70% of Internet users may speak English but are not native English speakers. This is largely due to Chinese, other Asians, Russians, Germans, Spanish speaking, Portuguese speaking, and Arabs, Iranians, and Africans that interact on the Internet within their language/cultural community. It is important to dispel the notion that communication on the Internet is in English. In its large majority it is not. But, global, intercultural communication is in English, as is, in fact, most international communication. The imbalance has consequences in the uneven access to resources on the World Wide Web. Figure 7.2 shows the relationship between the language of websites and the languages of Internet users, as of 2021/22.



**Note:** Websites as of February 2022; Internet users as of 2021.

**Source:** Statista, W3CTechs, Internet World Stats.

**Figure 7.2** Share of websites using selected languages compared to estimated share of Internet users speaking those languages

The gap becomes more significant when considering the relationship between the language distribution of the website and the share of Internet users speaking a given language. Overall, it means that 70% of users are unable to use their language on 60% of all websites. Since the knowledge of English is skewed toward the most privileged groups in every

non-English-speaking country, a fundamental cultural divide is introduced in a digitized world.

Granted, substantial progress in automated translation operated by AI allows us to overcome some of the language barriers. But this creates an additional inequality in terms of the skills necessary to operate and interpret IA programs.

The latest developments on AI build upon LLMs trained on massive datasets containing the text collected from the open web. As a result, applications like ChatGPT perform better with languages for which more websites are available. ChatGPT operates on databases that are mainly in English. For example, although it is estimated that there are 7,000 spoken languages and approximately 300 writing systems (UNESCO, 2018), ChatGPT claims it can respond in English, Spanish, French, German, Chinese, Japanese, “and many more,” but cannot establish a number to what others have estimated to be 95 languages (SEO AI, 2023). As a result, those who can speak popular languages will get access to state-of-the-art technologies, while others will be excluded from being able to take advantage of them. On the flipside, like the excluded proles of Orwell’s *1984*, they might be the only ones left with the ability to engage online freely, while those whose languages have been studied by machines will find their expressions constantly monitored and moderated by automated systems. During the world wars, the US leveraged their unique indigenous languages as part of a codification system the Japanese and Germans would not be able to break (CIA, 2008). Perhaps in the cyberwars of the future, the languages unknown to the machines will play a similar role.

Furthermore, languages are produced by culture, and our brains use language categories for their codification of experience. Therefore, beyond language translation we have to engage in a cultural decoding of our global communication capabilities, emphasizing the fact that we live in a global world, but we do not have, and will not have, a homogeneous global identity. “Citizens of the world” is just the self-aggrandizing definition of the global gentry (Castells, 1997).



## Meaningful access to the Internet

The United Nations 2030 Agenda proposes that Internet access should be universal and meaningful. The aim is 100% access for all adults, all schools, all businesses, with a mobile network of the latest technology, and universal ownership of a mobile phone by the adult population. Gender parity should be achieved, 70% of adults should have basic digital skills, 100% of fixed broadband subscriptions should be 10 Mb/s or faster, and 20 Mb/s should be the minimum download speed at every school. Entry-level broadband subscription should cost less than 2% of the average national income. The data presented in this chapter show how far most of the world still is from these targets. However, there is an even more fundamental question: what is the *meaning* of an Internet connection for people at large? A number of studies (Blank, 2013; Pew Research, 2013) have investigated the factors that lead people not to access the Internet. Old age is the most important factor. Indeed, in the US in 2021, 25% of people over 65 never go online. Low level in education deters potential users. Intensity of bonding and social capital also diminishes interest for Internet use, as does satisfaction with one's health condition. Gender, income, and occupation do not seem to predispose to this withdrawal from digital life. Thus, a potential portrait of the voluntary nonuser of the Internet emerges: individuals whose life cycle invites a retreat to their world of intimacy, and free from work-related pressures, most of them being retired; people in good health with strong social bonds who prefer selective person-to-person sociability, in other words, their meaningful communication is built outside the digital environment. On the opposite side of the spectrum, people with low education lack skills to find casual navigation of the Internet rewarding. Between the educationally excluded and those able to live life as it used to be, the overwhelming majority of the population needs digital networks to learn, work, and live, as citizens-netizens. It is out of the question not to use the Internet. So, they inhabit the Internet galaxy, where they meet the traditional sources of social inequality, and the uncertain quest for the meaning of their lives. The meaningfulness of the Internet does not result from any technological affordance. It depends on the meaning that everybody gives to their experience, thus adapting the new technologies to the fulfillment of their needs and desires.

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# 8

## Networked social movements

Social movements have always been the primary agents of social change. They aim at transforming what we think and what we value. Thus, they are cultural movements rather than political movements, those seeking to alter power relations in the State, although they often induce political effects as a result of the new values they put forward. They are formed by communication between people challenging the dominant norms enforced by institutions of society. Frequently they result from outrage provoked by acts of powerful actors that are perceived as unjust. Social mobilization follows as a protest against those acts. From these collective actions emerge ideas that are elaborated and debated in the movement, and ultimately may induce projects of social transformation, whose fate depends on the conflictive interaction between social movements and the institutions of society. Some movements become formal organizations. Most of them exist as loose forms of cooperation structured and focused in a communication process. How communication is established and which are the technologies of communication evolve historically and ultimately frame the social movements. In the twenty-first century, the advent of digital networked communication, as analyzed in Chapter 2, has induced a form of social movements characteristic of our time: networked social movements, whose features and dynamics I will present in this chapter. Networked social movements are not the result of digital communication. They respond to perceived oppression and injustice that are rooted in the dominant values and interests in a given society. Yet, the form they take, and their practices, are enabled by digital communication technologies. They evolve with technological change in interaction with cultural and political change.

## The early twenty-first century: a time of global change<sup>1</sup>

The sudden rise of networked social movements took place in the second decade of the twenty-first century, particularly after 2010, although before this date there were a number of powerful mobilizations facilitated by digital mobile phones in countries as different as South Korea (2002), Spain (2004), and Iran (2009).

However, the spread of these movements picked up in 2010 and subsequent years. In December 2010 similar movements first started on the Internet and then occupying urban space took place in countries as different as Iceland and Tunisia. The Tunisian revolution sparked what was labeled the Arab Spring in January 2011, with hundreds of thousands of people following calls on the Internet, motivated by outrageous behavior of the ruling elites. They occupied squares in the main cities in Egypt (Cairo's Tahrir Square became a symbol all over the world), Syria, Bahrain, Morocco, and smaller movements in other Arab countries. In Spain, the May-15th 2011 movement, protesting the management of the financial crisis and calling for "real democracy," started on the Internet and went on to occupy squares for over a month in most Spanish cities. Similar movements, with diverse intensity, took place in other European cities: Paris Debout, Lisbon, Athens, London, Amsterdam, Berlin. In Istanbul the Gezi Park movement challenged the municipal authorities. In Kiev, thousands of protesters opposed to the pro-Russian Government occupied Maidan square and prompted the Orange revolution in Ukraine, leading, years later, to a major impact on global geopolitics.

In the US, the Occupy Wall Street movement, started in September 2011, denounced the practices of financial speculation, and the protection of corporations by the government using public funds at a time when thousands of families were losing their homes because of mortgage defaults. The movement made the country aware of extreme inequality, with

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<sup>1</sup> Detailed empirical studies and sources on networked social movements 2010–15 are presented in Castells (2015). Subsequent Internet-based social movements in 2016–23 as recorded by academic research and limited ethnographic observation by this author have reproduced a similar pattern in their social practice, confirming the main lines of the analysis presented in this chapter.

1% of people controlling most of the wealth. “We are the 99%” was the slogan of the movement. From New York, occupations extended to Los Angeles, Oakland, Chicago, Atlanta, and to over 1,000 American cities with variable intensity. In Brazil, in June 2013 the youth movement for free transportation triggered a series of demonstrations against political corruption that shook up the Brazilian institutions. New movements in 2015 and 2016 showed that the elites had learned the new form of politics and mobilized the middle class to elect as president the ultrarightist Bolsonaro. However, in 2022, left-wing activists regained influence in the Brazilian digital networks and succeeded in just electing Lula, ousting Bolsonaro. In Chile, the student movement that started in 2012 was able to mobilize other segments of the population, leading in October 2019 to a series of occupations and demonstrations in defense of Dignity. As a direct outcome of this movement, in November 2021, Gabriel Boric, a young leader of the movement, was elected president. In Colombia, in April 2021, the “estallido social” (social explosion) took place against the mishandling of the health measures against the Covid-19 pandemic, and against murderous police violence on the demonstrators. In 2022, largely as a result of the “estallido social,” left-wing leader Gustavo Petro was elected president of Colombia and engaged in a determined peace process by negotiating with the guerrillas. In Mexico, social movements in the Internet, such as “#Yo soy 132” in 2012, were paramount in the defense of human rights and the exposure of corruption in the State. It paved the way for the election of populist president López Obrador. Less followed by the Western media, there were significant movements in other countries, in fact in over 100 countries. In Nigeria in 2020, the “End SARS” movement mobilized the population against a brutal special police unit. The government repressed the demonstrators and closed Twitter for 7 months as social networks had played a key role in mobilization. The youth activists of this movement came very close to electing president Peter Obi (Labour Party) in the 2023 election, although the usual political tricks derailed his probable victory.

In Hong Kong in 2014, the Umbrella Movement fought for democracy and was able to call massive demonstrations in spite of heavy-handed police repression. Even in China, a country where these movements had not appeared until then, in 2022 there were significant protests organized on the Internet in Shanghai and other cities against the confinement policies of the government during the resurgence of the Covid-19 pandemic.



In fact, these movements were able to reverse government policies, a rare case in China

In Iran in 2022, thousands of Iranian women, connected around mobile social networks, and supported by men, protested against the assassination of Mahsa Amini at the hands of the religious police. The movement was savagely repressed and some activists were hanged. Yet, the seeds of protest and the affirmation of new values remain in the minds of Iranian women. In Israel in 2023, hundreds of thousands marched against the violation of the constitution by the Netanyahu government. In France in June 2023, the French “banlieues,” where ethnic minorities are segregated on the periphery of major cities, revolted violently for over one week, protesting the assassination by the police of 17-year-old Nahel, and calling for an end to institutional racism, prevalent in the police.

Thus, networked social movements cut across ideologies, issues, or institutions, as they become the global form of social change in our time.

## **Anatomy of networked social movements**

Networked social movements are highly diverse around the world. Yet, they share some fundamental characteristics in spite of substantial differences between the cultural and institutional contexts in which they happen. Indeed, it is because of this similarity in spite of contextual differences that I suggest that we are in the presence of a distinctive form of social movement that emerges from the organizational features and communication technologies underlying our social structure on a global scale: the network society (Castells, 2000). Observation of these social movements by a plurality of researchers, including myself, shows a recurrent pattern, with slight variations depending on specific settings (van de Donk et al., 2004; Earl & Kimpot, 2011; Castells, 2015; Carty, 2018; Jackson, 2018; Brown et al., 2022). I will summarize below the main components of this common pattern and the conditions of its interaction with the social structure and institutions where collective action aimed at social change takes place in contemporary societies.

First of all, their organizational form and their practices are based on networks: social networks and communication networks usually enabled

by digital communication technologies, particularly mobile communication devices connected by the Internet. Since the smartphone was first introduced in 2007, it appears probable that this technology made possible an exponential increase of social networks in the practice of social movements at the same time that the networking logic diffused in all domains of human communication and organization. Indeed, while mobile phones played a significant role prior to 2007 in some social protests, the growth of Internet-based communication networks allowed the explosion of these movements after 2010.

Second, although the large majority of these movements formed through interactions on the Internet, they became socially meaningful by networking people in the urban space, be it in street demonstrations, as in Brazil, Syria, and Iran, or by occupying squares and buildings for a period of time, as in Egypt, Spain, the US, and Ukraine. The spatial expression of the movement, regardless of its duration in time, is essential to make it visible to the entire society, beyond those connected on the Internet. Furthermore, the physical connection enhances the feeling of togetherness that is a source of strength necessary to face the fierce police repression that will inevitably hit the movement. In fact, the simple fact of illegally occupying space challenges the powers that be, and materializes the protest in a form that makes it easy for people to join the movement, just by going to the occupied space without requiring a formal membership.

Thirdly, the combination of networked interaction both in cyberspace and in urban space constructs materially and culturally a space of autonomy, an essential feature of social movements in all historical instances, as this autonomy is required to freely socialize, deliberate, and decide outside the reach of the established procedures of the society that are being challenged. Looking back in history, the industrial working-class movement in Britain emerged from the togetherness felt in the large factories, in combination with spaces of autonomy such as the British pubs (Thompson, 1963). The Commune of Paris in 1871 and multiple movements of alternative projects of common life, in the utopian or anarchist traditions, played a similar role (Castells, 1983). The space of autonomy in the network society is built around the Internet networks and in the occupation of urban space, however brief this may be.

This form of autonomy explains the fourth and most essential feature of these movements: they are rhizomatic. When the institutions of society (judicial authorities, police, and the like) force the movement to retreat, overwhelmed by coercion, more often than not they do not disappear: their values, their projects, as well as many of their actors take refuge in the Internet, and they surge again when the reproduction of social injustice at the roots of the outrage that gave birth to the movement recur. The rhizomatic feature of these movements helps their survivability. While Internet networks can be surveilled, and some rare times even shut down, our societies cannot operate without the Internet and digital communication networks, and so any dormant seed of resistance can be activated at any time, although with different themes and projects.

Fifth, networked movements do not need formal organization or established leadership. The network is the message. Its nodes and their relative hierarchy evolve with the movement and its interaction with the institutions. Membership is loose, discipline absent, replaced by influence and self-organization. When the movement evolves to involve political actors with specific programs, then formal organization and leadership become essential. But this is a different kind of social actor whose origins may be in the social movements but whose logic and practice are substantially different. They are political actors, seeking political power by acting on the State. Social movements, in contrast, explore and propose values, blueprints of life, acting on the minds of the people.

Lastly, social movements are rarely programmatic. They put forward demands and refer to values and principles, but they do not propose a realistic set of institutional measures. In fact, in many cases the movement anticipates in its practice what would constitute its desirable forms of institutional organization. For instance, the frequent goal of achieving “real democracy” is exemplified in the practice of deliberative democracy, debating and deciding in open assemblies in the squares and in consultations on the Internet. In other instances, social revolts propose values in their practice without making them explicit. For instance, the Black Lives Matter movement in the US in 2021, or the violent protests in the French “banlieues” in 2023, shifted from denouncing police profiling and unchecked brutality to the ideal of eradicating racism from the institutions. This is a fundamental value to ensure peaceful coexistence in our multiethnic societies that is implicit in the insurgency against the systemic police repression that targets ethnic minorities. To end racism would be

a major cultural and political revolution. However, while movements are not programmatic *stricto sensu*, there are certain values that appear consistently in the discourse of many movements. The most important of these common values, across cultures and countries, is the defense of Dignity, a term repeated in different languages, from Madrid to New York, and from Kiev to Santiago de Chile. People mobilized to be recognized and respected as human beings, regardless of their gender, age, ethnicity, education, income, or legal status. It would appear that beyond the specific sources of the protest and the goals of the mobilization, most of these actions are movements for human rights, encompassing all the claims for equal treatment to people by the institutions beyond the narrow definition of citizen rights. Dignity is the value claimed over the entire world by movements that aspire to expand equality and liberty to all domains of human experience.

Social movements always fade away, either obliterated by repression (including cultural repression in and by the mass media), or co-opted and institutionalized in their values and in their actors. Yet, the forms of their death are not irrelevant. They may die in an apparently useless effort to erode the fortresses of society. But also, in many cases, they survive and blossom in terms of the ideas and ideals that they project, out of the outrage that prompted their existence. They can induce hope for a better life and a better world in the minds of people, ultimately enshrined in the institutions. This is why they are so deeply misunderstood by media and politicians, always obsessed with tangible results. In fact, their potential outcome is the transformation of values and institutions of society. They are the salt of the Earth.

## **Social movements and communication technologies**

Social movements are a permanent feature of human history. They display multiple forms, but are always based on a process of communication by which outrage is shared, demands and values bring people together, and collective action is organized. This is why the control of information and communication was always paramount to preserve power. Consequently, only by breaking the hold of the powerful elites over communication could social change occur. This could be the work of prophets, of oral diffusion of criticism and new ideas, of communicable texts, and ulti-

mately of pamphlets, books, and the print press. This is why books were banned or burned, and reading and writing was confined to the elites for centuries, and censorship, by religious or political authorities, was and is a decisive lever of power making (Castells, 2009). With the advent of digital social networks communicated over the Internet, the monopoly of mass communication was challenged. Not that power dissolves, because it is present in the interaction in the social networks, and because corporations own the networks and governments retain regulatory powers, as I have analyzed in previous chapters of this volume. Yet, full control of information becomes more difficult, and the communicating actors have a fighting chance to connect, share outrage, counteract manipulative messages, and elaborate alternative projects.

Social protests, eventually developing into social movements, are not the result of digital networks of communication. They surge from the pits of suffering of everyday life. But because our society is largely based on these networks, in every domain, their logic and their affordances extend to all human practices, including social movements. This is why in the network society social movements take the form of networked social movements. They could not exist in their current form without the digital communication paradigm. Technological foundations of collective action do not prejudge the values or projects of the networked actors. Social mobilizations aiming at the restoration of the traditional social order coexist with movements seeking social change. In both cases their logic is shaped by the specific characteristics of digital technologies.

Digital communication is based on networks, and this is the form that practically all social protests and social movements take in our time. What follows is the increasing survivability of networked social movements, in terms of their rhizomatic dynamics and their resilience to targeted repression. They often become instant “communities of practice,” without the need for formal organization and leadership, because their coordination and deliberation can always happen over the Internet. Communities of practice are increasingly important forms of social organization in all domains of society, from business to science and education (Hughes et al., 2013). They also characterize social movements, transforming their dynamics and making possible horizontal patterns of coordination and guidance.

Outrage at an action by a powerful social actor has always been the most frequent stimulus to spontaneous protests that often overwhelm the repressive capacity of the established institutions, forcing at least a negotiation to preserve social order under new regulations. Shared outrage continues to be the spark that lights the fire of social revolts in our society. The specificity of our contexts is the capacity to visualize and distribute images of the outrageous behavior. The influence of images in our minds is the most important trigger of emotions. And emotions are the main determinant of collective action. YouTube, Instagram, Twitter, and TikTok, among other social networks, have been the most decisive tools to ignite social protests. Furthermore, the pervasiveness of smartphones makes it possible for anyone to record an outrageous act (such as an arbitrary killing by a police officer or the torture of animals) and immediately upload it to the Internet.

Virality is a fundamental feature of Internet communication (Nahon & Hemsley, 2013). Virality makes possible the almost immediate distribution of a message to a very large audience. The quasi-simultaneity of the reception of the message helps the spontaneity of the protest, by easy, fast feedback between communicating subjects who, in their interaction, scale up from individual emotion to collective mobilization. The new technologies of connectivity (5G, 6G), besides increasing the speed of communication, reduce the latency of the response, facilitating the interaction of outraged individuals in real time.

In sum, digital communication technologies, combined with imaging techniques, and the uses of artificial intelligence in constructing images, shape social movements caused by reactions against perceived social injustice in ways that empower considerably the new actors of the process of social change.

## **Networked social movements and political change**

While the fundamental outcome of social movements refers to changes in our way of thinking and being, with different timing depending on contexts, the most immediate traces of their potential as agents of change appear to be in their impact on the State, as the direct expression of power relations. The empirical record on this issue shows a wide range

of outcomes. In most cases of the observed practices in the second decade of the twenty-first century, the political outcomes have been highly significant. However, not necessarily by fulfilling the goals of the movement. In fact, in many instances the powers that be reacted with extreme brutality and crushed the movements, sending dark waves of despair across the world. The Arab Spring did topple decades-old dictatorships in Tunisia and Egypt, and challenged Assad in Syria. Yet, Tunisia ultimately became a corrupt pseudo democracy, the Army took control over Egypt, supported by the US, and Russia came to the rescue of Assad. Then, Al Qaeda, Saudi Arabia, Iran, Israel, and the Western powers joined the fray, triggering an atrocious war in Syria with hundreds of thousands of deaths and millions of refugees.

Social movements disappeared because they are always the first victims of wars. Ideas may prevail in the long term. But missiles, tanks, and terror overwhelm ideals in the short term. Nonetheless, I dare to consider these catastrophic political effects as being induced by the social movements of the Arab Spring. Negative effects are, analytically speaking, as important as positive ones. In the US, the Occupy Wall Street movements planted seeds for the crisis of elite legitimacy in the minds of millions of citizens. For instance, they became fully aware of the extent of social inequality contradicting the American dream. Institutions were delegitimized, including political parties, financial elites, and the media. Paradoxically, the main political outcome was the rise of right-wing populism, exemplified by Trump and his Make America Great Again (MAGA) ideology reaching the presidency against all odds, thus changing American politics and ultimately endangering democracy. This is because the challenges that emerge from social movements have to be processed in the political system, and when there is no relay between alternative projects for society and the political actors, opportunistic demagogues tap on the nostalgia of better times that never were, instead of exploring new forms of living together.

However, in several countries some social reforms and political democratization were implemented as a partial response to social movements. In Latin America, social movements resulted in major progressive political changes in Chile, Colombia, and Mexico, in spite of the limitations of left-wing populism. Spain offered perhaps the most direct example of how powerful social movements led to the emergence of a new political actor (Podemos) and to a new left-wing orientation of the powerful Social

Democratic Party. Together they formed the only center-left coalition government in Europe, coming to power in 2020, at the very time that the Covid-19 pandemic hit the world and the country. They successfully managed the health situation and the economy, and they enacted a number of progressive environmental, feminist, and social policies. Only to be hit again, in 2022, by the Ukraine war and the war-induced inflation. Yet, they were reelected in 2023 in alliance with Catalan, Basque, and Galician nationalists, making possible a multinational coalition in the Spanish State for the first time in history.

This is to say that social movements, by bringing new values to the public agenda, may change the political scene and government policies, but political outcomes will depend on the interplay between political actors and political dynamics in society at large. And societies are usually prisoners of fear and averse to profound changes. So, from outrage to hope to political change, there are a number of intermediate institutional layers, specific to each society, that refract and modify any new project of life created by the social movements. In this interplay between social change and social resistance to change, communication networks are essential as they process the ideas and will that configure the minds of citizens. Networked social movements interact with networked informational politics (Castells, 2017).

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# 9

## Social media and political polarization<sup>1</sup>

In all societies and throughout history, power relationships are the producers of norms and institutions of governance. Power relationships are by and large played out in the realm of communication (Castells, 2009). This is because the neural networks of human brains communicate with the brains of other humans and with the networks of their natural and social environment. Humans are social animals; thus, they live and act by their communication networks (Lakoff, 2008). From this interaction results the forms of social organization and institutions in which power relationships are embedded. The stable reproduction of these relationships takes place in the political process, which frames the competition for power positions in society. Institutional stability depends on the acquiescence of subjects of a given social system to respect the interests and values of actors who occupy power positions in the institutions. In this case, the institutional system is perceived as legitimate, or inevitable, by a sufficient majority of citizens. Yet, all institutions are in flux as actors, values, and interests that are in a subordinate position try to improve their standing by negotiating or challenging the norms enforced by the State. When the challenge from social actors reaches a certain level of intensity, the political institutions and their smooth operation suffer a crisis of legitimacy. Namely, large segments of society reject their authority.

A key factor in this potential crisis of legitimacy is the social dynamics taking place in the space of mass communication—a space that, as I explained in Chapter 2, depends on the characteristics of the senders of messages that could reach a mass audience, and on the technological medium they use. This space is organized and regulated, and becomes a critical component for the assertion of political legitimacy as well as for its potential crisis.

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<sup>1</sup> The analysis presented in this chapter does not apply to the use of social media in authoritarian regimes, as polarization is suppressed by political censorship.

For most of human history, the technology of mass communication made possible the formation of organizations and institutions that were in control of the communication process, be it by the State, by economic organizations (e.g., corporate business), or by ideological apparatuses (e.g., churches or religious institutions at large). With the rise of networked digital communication technologies, this control is weakened. People-to-people networks can now reach an interactive, horizontal form of mass communication, disintermediating the powers that be. However, this techno-utopia is somewhat belied by the appropriation of these new platforms of digital communication by the new forms of capital and the State, the old nemeses of unfettered communication.

Yet, since communication controls are more difficult to enact in the digital paradigm, and since the new masters of communication have vested interests in stimulating communication within their proprietary networks, the scale and diversity of horizontal communications is such that relative communicative autonomy may challenge the systemic domination of established values and interests. This is why the traditional political institutions usually blame the vitality and diversity of social media communication, in spite of their oligopolistic appropriation, as the source of the crisis of legitimacy they suffer all around the world. In their view, this new trend results in an increasing ideological and political polarization that shakes institutional stability, ultimately threatening the very survival of liberal democracy.

Society *is* increasingly polarized. This polarization occurs in multiple domains of social life as major conflicts shake out the common ground of beliefs. This can be seen in the prevalence of sexism, homophobia, racism, xenophobia, religious fanaticism, radical nationalism, ethnic hatred, according to data from reliable sources (Achieme, 2018; OHCHR, 2019; Khan, 2021; World Health Organization, 2021; Sardinha et al., 2022).

Each one of the expressions of nonnegotiable antagonism has specific historical, cultural, and social roots. But what characterizes the current polarization is the fading away of trust in the institutions and leaders of all countries. Without trusted mediating institutions able to manage conflicts, the clash in values and interests appears to be irreconcilable. The end of trust ushers in an era of uncertainty. But what is the specific role of social media in the process of polarization?

A stream of scholarly research in recent years shows a complex picture of the ideological transformation underway. There is no doubt that the legitimacy of established institutions has been substantially eroded in the twenty-first century (Castells, 2009, 2015, 2017), not just concerning the State, the political system, and the political actors. It also refers to the mainstream media, the main economic actors (particularly the financial system) as well as traditional religions, official culture, and even science—as apparent by significant resistance to the vaccines that saved our lives during the Covid-19 pandemic.

In my book *Rupture: The Crisis of Liberal Democracy* (2017), I documented the extent of the legitimacy crisis worldwide, starting with the lack of trust of two-thirds or more of citizens in political parties and political leaders that govern their societies. Distrust of the elites and of their messages has dramatically increased since then in democratic countries (Edelman, 2022), partly as a consequence of the anger provoked by the confinement measures imposed to control the pandemic. Key factors in inducing the crisis of legitimacy appear to be the feeling of unfairness derived from rampant social and economic inequality, the resentment of government's systemic support of oligopolistic corporations, the widespread bureaucratization and perceived corruption of political parties and their leaders, and the shrinkage of the welfare state after the policies of austerity that followed the 2008–12 financial crisis.

Other factors contribute to the downfall of public trust: fear of fast, uncontrolled technological change, the backlash against the challenge to patriarchal domination, and their corollary: pervasive violence in women's everyday lives. Moreover, the growing awareness that climate change threatens the survivability of humans on the blue planet deepens uncertainty about our future.

Altogether, a perfect storm is forming with nobody at the helm of our fragile vessel of survival because of the distrust of institutions and elites. Collective uncertainty and individual anxiety populate the Internet, spreading rumors, disinformation, and misinformation under multiple disguises. We live in a post-truth age, and a key component of this misinformed bewilderment is the distortion of reality in a mass self-communication system with little regulation. Because the mass self-communication sphere that comprises billions of daily users is largely in the hands of private oligopolies whose business model is based

on increasing traffic and generating data that they can capture with little government or judicial control (Taplin, 2023). Data capitalism is one of the most dynamic sectors of informational capitalism, as I analyzed in Chapter 3. The outcome of such a multidimensional transformation of communication is an immense cacophony of segmented, mass communication that mixes truth, lies, and partial lies, in a cloud of special effects and symbolic manipulation operating at light speed, saturating our lifetime, colonizing our personal space, thus blurring time, space, and experience.

Yet there is something else: is our daily practice of social media and their pervasiveness the primary source of the ideological and political polarization that is making it increasingly difficult to live together (Touraine, 2012)? These are the kinds of questions for which social science could provide some answers without yielding to the temptation of just asserting a subjective preference. The answer, that I will try to support empirically, is *yes* and *no* at the same time.

The bulk of political and ideological expressions in social media chat and publishing outlets are definitely polarized. However, what we would normally classify as political and ideological exchanges are relatively few in number when considering the scale of people participating within these sites. True, anyone with an opinion can post practically anything without yielding to any social norms of respect vis-à-vis dissenting opinions (Persily & Tucker, 2020).

Furthermore, most people gravitate to social networking sites, which includes specific groups within a platform, that reflect their own views, and toward mass media broadcasters that will confirm their views. This is because our brains tend to discard information that challenges any deep-seated perceptions and opinions (Castells, 2009: 167–169). A number of studies have shown, for instance, that, in the US, the audiences of Fox News and of National Public Radio do not overlap at all (Arsenault & Castells, 2008). In fact, the business model of Fox News aims at controlling the audience of ultraconservative viewers. The same logic applies within social media.<sup>2</sup> Groups within social media platforms are formed by affective affinity (Bakshy et al., 2015). The formation of

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<sup>2</sup> It is interesting to look at the ideology of people who financially supported Elon Musk in his bid to take over Twitter, a company playing a role in

attitudes and the projection of behavior take place in what is known as “echo chambers,” where the interacting subjects are rewarded by confirming their opinions and elaborating their views on a shared value system (Barbera, 2020). Thus, the polarization that exists in society determines the polarization in the mass media and in the media.

What is the role of media interaction with these pre-established opinions? It depends on how social media sites construct the interaction. The seminal work by Tucker et al., analyzing the empirical evidence on the matter, highlights three fundamental findings (2018):

1. Increased media fragmentation that mirrors the cultural and political fragmentation of society, replacing political news with entertainment, which ultimately lowers the quality of the political information received in the media space.
2. On the other hand, those people who do consume political information in the social media at large are exposed to a greater diversity of opinions, which may increase their civic engagement.
3. The exchange of explicitly political views in social media sites is frequently negative and uncivil, contributing to affective political polarization.

Therefore, the use of social media for all kinds of purposes may in fact increase civic consciousness and tolerance, while participation in partisan social media sites and conversations exacerbates polarization. Barbera, discussing a substantial body of research on the matter, finds that: “It is criticism of partisan identities, and not necessarily online opinions on specific issues that drives polarization” (2020: 45). In this regard, I would like to point out a crucial observation that has influenced the conclusions of many researchers: according to the study by Boxell et al. (2017) “greater Internet use is not associated with faster growth of political polarization among US demographic groups.” Indeed, while between 1996 and 2016 political polarization increased substantially in all age groups, it increased twice as much among people older than 65, than among those aged 18–39, while the younger population is the most active group in using the Internet. As they write: “We find that polarization has increased the

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a sector where most CEOs and employees are more aligned with the values professed by the Democratic Party.

most among the groups least likely to use the Internet and social media” (p. 10612).

However, for people who are already politically polarized, the use of partisan platforms and social media groups exacerbates their radicalization, particularly among those on the extreme right. It deepens polarization and intolerance in society at large because they are the most visible in the communication space (Nielsen & Fletcher, 2020). Four mechanisms seem to be operating to induce the polarization effect. The first is the addictive quality of intense engagement in social media controversies. As mentioned in Chapter 2, social media platforms are designed to keep users on their platforms for as long as possible. The timing of the delivery of “likes” and comments from followers and users sharing the same views, as well as the negative reaction to comments from opposite views, is leveraged and manipulated to produce a rush of dopamine, rewarding the brain with a feeling of excitement and satisfaction. Thus, the more people are engaged in social media activity, the more the dopamine neurons are active. And the anticipation of pleasurable effects increases participation in the social media that support this activity in a recurrent feedback loop (Burhan & Moradzadeh, 2020).

Secondly, the bias toward (mis)information that fits the position of the most engaged people contributes to the spread of such misinformation (Wittenberg & Berensky, 2020).

Thirdly, virality speeds up the diffusion of misinformation that is accepted uncritically by the users that are predisposed to believe certain contents, and so they quickly forward the messages.

Fourthly, in the case of targeted social media campaigns seeking to impact opinions, the massive use of programmed (ro)bots allows the distribution of the message to be amplified exponentially and reach a vast audience that could easily be overwhelmed by exposure to a myriad of messages tailored to the characteristics of the receivers.

Under such conditions, any attempt to counter the misinformation by posting reminders of basic facts is disadvantaged by the sheer size of the mass of misinformation. Subjectivity cancels objectivity. The massive use of bots in the forwarding of the messages is widely considered to have been a critical factor in the electoral victories of Trump in the US

(2016), Bolsonaro in Brazil (2018), the Brexit campaign in the UK (2016), Meloni in Italy (2022), as well as in the growing influence of such extreme right groups as Front National in France, Vox in Spain, and the Swedish Democrats.

It needs to be emphasized that these campaigns are very expensive and require resources and sophistication that only major business groups or powerful governments have at their disposal. Such was the case with the support of the American Koch brothers for Bolsonaro and of the Russian Government for Trump, and probably also for Le Pen and Meloni. In other words, this is not a blind automated process enacted by social media platforms, but a political strategy that skillfully uses the potential of social media and the vulnerability of the active users. The more people are engaged and the more they are susceptible to messages that they “like,” then the more social media is influential.

The media platforms have an incentive to stimulate a polarized debate since it increases the traffic. The more users become active in the defense of their ideas and perceptions, and the more they are mobilized, positively or negatively, in the social media networks, the more data that can be captured or created.

When companies yield to the pressures of governments or of public opinion, they moderate some of the most extreme expressions of uncivility and misinformation through removal or a downranking of visibility in the algorithmically curated feeds. Complete removal through content takedown is extremely limited. For instance, in 2022, TikTok announced it removed over 110 million videos during April to June 2022, which accounts for less than 1% of the total number of videos posted during that period (TikTok, 2022). We should always keep in mind that the business model of most companies managing the digital platforms is based on increasing traffic in the networks, any kind of traffic. For this reason, they tend to prefer relying on curation that manages what content is most visible and for whom rather than full removal.<sup>3</sup>

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<sup>3</sup> For example, in 2018 Zuckerberg released a blueprint on content moderation where he acknowledges the conflicting incentives Facebook is subjected to: “One of the biggest issues social networks face is that, when left unchecked, people will engage disproportionately with more sensationalist and provocative content. This is not a new phenomenon. It is widespread on



And so the loop is closed: ideological and political polarization in society is concentrated in the activity of vocal minorities that hold on to their beliefs in dedicated spaces and relentlessly fight opposing views. In so doing, they contribute to further polarization in society, while enhancing the power and gains of the platforms by increasing user activity and providing a wealth of data that are appropriated and marketed by the owners of the networks. The technology masters of the world link up with oligopolistic business groups to establish profitable alliances with the political leaders of society, very often regardless of their ideological positions. There are, however, new instances of resistance to alternative projects and values enacted by social actors that are increasingly active in the public debate. Yet, by doing so, they increase further interaction and traffic, helping the accumulation of data for capitalism. At the same time, they broaden and intensify public debate, which often includes misinformation, that may undermine the political legitimacy of the power holders across the political spectrum. The political actors that are winners and losers in the debate may vary, but over time the legitimacy of the political system as a whole is eroded.

In sum: social media do not create political and ideological polarization, which has its roots in the conflicts within society. But they exacerbate and broaden polarization, using the digital public sphere as an amplifier of uncivil ideological confrontation.

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cable news today and has been a staple of tabloids for more than a century. [...] Our research suggests that no matter where we draw the lines for what is allowed, as a piece of content gets close to that line, people will engage with it more on average—even when they tell us afterwards they don't like the content. [...] This is a basic incentive problem that we can address by penalizing borderline content so it gets less distribution and engagement" (2018).

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